

Envelope bounds for the canonical coefficients

Claude, with Daniel Murfet

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Abstract

Towards a uniform Taylor tree. Under a Gaussian envelope $|c(s)| \leq K(1+s)^D e^{\beta s a'}$ the log moments obey $|M[\alpha; \ell; c]| \leq K \int_0^\infty s^{\alpha-1} (1+|\log s|)^\ell e^{-\beta s^2} (1+s)^D e^{\beta s a'} ds$ (`abs_logMoment_le_of_env`); a discarded pole term obeys $|N^{-\alpha}(A \log N + B)| \leq (|A| + |B|)N^{-2T}(1+\log N)$ for $\alpha \geq 2T$, $N \geq 1$ (`discard_explicit`); and for the analytic amplitude with envelope (C_0, L, D, ρ) the canonical coefficient functions $U_\alpha, V_\alpha, C_\alpha$ have explicit envelopes (`canonUEnv`, `canonVEnv`, `canonCEnv`), whence $|A_\alpha| \leq \text{canonCEnv}(\alpha) EM(\alpha, 0)$ and $|B_\alpha| \leq \text{canonUEnv}(\alpha) EM(\alpha, 0) + \text{canonVEnv}(\alpha) EM(\alpha, 0) + \text{canonCEnv}(\alpha) EM(\alpha, 1)$ (`canonA_abs_le_env`, `canonB_abs_le_env`), all constants depending only on the envelope data. Zero sorry/axiom.

1 The things formalised

```
theorem abs_logMoment_le_tailMoment ( : ) ( : ) (c : → ) (hc : Measurable c)
  (hint : IntegrableOn (fun s => s ^ ( - 1) * (1 + |Real.log s|) ^
    * (Real.exp (- * s ^ 2) * |c s|)) (Ioi 0)) :
  |logMoment c| tailMoment c
theorem abs_logMoment_le_of_env ( a' K : ) ( D : ) (c : → ) (h : 0 < )
  (h : 0 < ) (hc : Measurable c) (henv : s, 0 < s → |c s| K * gaussEnv a' D s) :
  |logMoment c| K * envMoment (gaussEnv a' D)
theorem discard_explicit (T A B N : ) (h : 2 * T ) (hN : 1 N) :
  |N ^ (-) * (A * Real.log N + B)| (|A| + |B|) * (N ^ (-(2 * T)) * (1 + Real.log N))
noncomputable def canonUEnv (b C : ) (h h k k : ) ( : ) : :=
  if uExp h k (uIdx h k ) = then
    fpEnvConst b k (canonicalM ) (C * ( ^ i) ^ 1 * (1 + ^ i) ^ canonicalM )
    (C * ( ^ i) ^ 1 * ( ^ canonicalM ) ^ 1 / (1 - b / ))
    -- = k - h - 1, i = uIdx
  else 0
noncomputable def canonCEnv ( C : ) (h h k k : ) ( : ) : :=
  if uExp h k (uIdx h k ) = vExp h k (vIdx h k ) = then
    C * ( ^ (uIdx h k + vIdx h k )) ^ 1 / ((k : ) * k)
  else 0
theorem canonA_abs_le_env ... ( : ) (h : 0 < ) :
  |canonA h h k k (fun i j s => c (i, j) s) |
  canonCEnv C h h k k * envMoment 0 (gaussEnv L D)
theorem canonB_abs_le_env ... ( : ) (h : 0 < ) :
  |canonB b h h k k (anaFaceU c b) (anaFaceV c b) (fun i j s => c (i, j) s) |
  canonUEnv b C h h k k * envMoment 0 (gaussEnv L D)
  + canonVEnv b C h h k k * envMoment 0 (gaussEnv L D)
  + canonCEnv C h h k k * envMoment 1 (gaussEnv L D)
```

2 Mathematics

$|\int f| \leq \int |f|$ and $|\log s|^\ell \leq (1 + |\log s|)^\ell$ give the tail-moment bound; the pointwise envelope then gives the Gaussian-moment bound (unit 132). The canonical u -face coefficient at a u -pole is $k_1^{-1}\text{FP}_\gamma$ at the canonical truncation order, so its envelope is the finite-part envelope of unit 132 with the analytic adapter’s Taylor-data and remainder envelopes at that order; the collision coefficient $c_{ij}/(k_1 k_2)$ has envelope $C_0 \rho^{-i-j}/(k_1 k_2)$. Off the poles the coefficients vanish and the envelope constants are set to zero.

3 Paper-facing meaning

This provides the missing uniform bound on the discarded pole terms of the Taylor tree (Astra #9: “`discard_isBig0` needs uniform coefficient bounds”), so that the “for every T ” statement can be made uniform over the analytic family with a constant K_T depending only on the envelope data and T . It also bounds the canonical coefficients themselves in terms of the envelope, the first half of Astra’s R6 (coefficient maps bounded on the weighted family).

4 Lean notes

- **Name clashes are invisible to the single-file check.** `lake env lean` on the new file sees only its import closure; a lemma name already used in another `Laplace.Grammar` file outside that closure fails only at `lake build` (“environment already contains”). `grep` the namespace for the intended name before committing to it.
- `norm_integral_le_integral_norm` is the Bochner form of $|\int f| \leq \int |f|$; convert the goal’s absolute value with `← Real.norm_eq_abs` first, and provide integrability of $\|f\|$ via `Integrable.mono'` against the tail integrand.
- `split_ifs` on a definition whose branches are functions (`fun s => ...` vs `fun _ => 0`) works after `unfold`; the zero branch closes with `simp`.
- `positivity` cannot prove $0 < 1 - b/\rho$; a two-line lemma (`sub_pos`, `div_lt_one`) is worth having.